

independent implementation by providers. Runtime instances of the JVM are created when a user runs a Java class using the java command on the command prompt.

In summary, the JDK is a development environment that includes tools for creating Java applications, the JRE provides the minimum requirements for executing a Java application, and the JVM is responsible for executing Java bytecode.

Let's start by clarifying the differences between JDK, JRE, and JVM.

1. JDK (Java Development Kit) is a software development kit that provides the environment for developing and executing Java programs. It includes both the development tools (for creating Java programs) and the JRE (for executing Java programs).
2. JRE (Java Runtime Environment) is an installation package that provides an environment for running Java programs on your machine. It includes the JVM and some core libraries required to run Java applications.
3. JVM (Java Virtual Machine) is a crucial component of both JDK and JRE, as it is built into both of them. Whenever you run a Java program using JRE or JDK, it is executed by the JVM, which interprets and executes the Java code line by line.

Let us explore the components of JRE in order to better understand its importance and how it works. The following are the components of JRE:

1. Deployment technologies, which include Java Web Start, deployment, and Java Plug-in.
2. User interface toolkits, which include Swing, AWT, Accessibility, Java 2D, Image I/O, Sound, input methods, drag and drop (DnD), and Print Service.
3. Integration libraries, which include JDBC, Java Naming and Directory Interface (JNDI), Remote Method Invocation (RMI), Interface Definition Language (IDL), scripting, and Remote Method Invocation Over Internet Inter-Orb Protocol (RMI-IIOP).
4. Other base libraries, which include serialization, international support, Java Native Interface (JNI), Java Management Extensions (JMX), Networking, Override Mechanism, Security, I/O, Java for XML Processing (XML JAXP), and extension mechanism.
5. Lang and util base libraries, which include versioning, management, zip, reflection, instrument, Collections, Regular Expressions, Preferences API, Java Archive (JAR), Concurrency Utilities, Logging, and Ref Objects.
6. JVM which includes both Java HotSpot Client and Server Virtual Machines. JVM is a runtime interpreter that becomes an instance of JRE during the